

# claude-windows / a9fd4426-f234-4e4a-bfb4-1f2c8255c661

## Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-muneem\9fd4426-f234-4e4a-bfb4-1f2c8255c661\subagents\workflows\wf_076beba8-59c\agent-ac239473e2fb38507.jsonl`  
- SHA-256: `2c5a71e5ceda1e02d0350ffe88807d6ac4e343d75da8310a3047f252c6cae414`  
- Source modified: `2026-05-31T04:25:43+00:00`  
- Imported at: `2026-07-05T16:48:20+00:00`  
- project: `wf_076beba8-59c`  
- session_id: `a9fd4426-f234-4e4a-bfb4-1f2c8255c661`
```

## Transcript

1. user (2026-05-31T04:24:36.629Z)

### Adversarial Claim Verifier (voter 3/3)

Be SKEPTICAL. Try to REFUTE this claim. 2/3 refutations kill it.

### Research question

Identify the best LOCAL (self-hosted, single NVIDIA CUDA GPU) vision / document-AI / OCR models ? and, as a clearly-flagged approval-gated fallback only, paid cloud API services ? for parsing BANK STATEMENT PDFs into structured transaction tables (date, narration/description, withdrawal/debit, deposit/credit, running balance), for a LOCAL-FIRST personal-finance tool ("muneem").

Target documents: Indian bank statements (HDFC, Axis, ICICI, AU Small Finance) ? dense, multi-column, often SEMI-RULED or borderless tables, sometimes multiple accounts per statement; both born-digital (text layer present) and the harder cases where pdfplumber's extract\_tables/coordinate clustering is unreliable (e.g. Axis, whose column layout varies between statements).

HARD CONSTRAINTS (a model that violates these is unusable for us):

1. MUST run 100% LOCALLY for statement contents (highly sensitive). NO cloud APIs by default. Paid/cloud is acceptable ONLY as an explicitly-approved fallback ? still report the best cloud options but label them clearly as "cloud/sensitive ? approval-gated, not default".
2. MUST be usable from Python (transformers / vLLM / Ollama / ONNX Runtime / native lib).
3. License MUST be permissive & commercially safe for BOTH the CODE and the MODEL WEIGHTS: MIT, BSD, Apache-2.0, ISC, HPND. AVOID and explicitly FLAG anything GPL/LGPL/AGPL, CC-BY-NC / non-commercial, research-only, or custom "community"/acceptable-use licenses with restrictions. This is critical ? many document-AI models and their training datasets are non-commercial (e.g. LayoutLMv3 weights are CC-BY-NC; Surya and Marker use GPL-or-commercial dual licensing; several VLM weights ship custom restrictive licenses). Verify the WEIGHTS license, not just the repo license.
4. Hardware: a single consumer/prosumer NVIDIA GPU. Give VRAM tiers (~8-12 GB consumer, ~16-24 GB) and which models/quantizations fit each.

Evaluate and compare these LOCAL approaches ? for each give: code license, MODEL-WEIGHTS license (state explicitly), latest release / maintenance status as of 2025-2026, VRAM need + quantization options, Python integration path, and reliability specifically on DENSE FINANCIAL TABLES / multi-column statements:

- Open vision-language models (VLMs) for document/table understanding: Qwen2-VL and Qwen2.5-VL (note per-size license ? which sizes are Apache-2.0 vs restricted), MiniCPM-V 2.6, InternVL2 / 2.5, Microsoft Phi-3.5-vision, Mistral Pixtral, Meta Llama-3.2-Vision (flag the Llama Community License restrictions), Moondream2, GOT-OCR2.0, olmOCR and other OCR-specialized VLMs.
- OCR-specialist / document-AI pipelines: PaddleOCR + PP-Structure (table recognition), docTR, EasyOCR, Tesseract (already used), Microsoft Table Transformer / TATR (check code AND PubTables-1M dataset/weights license), Donut, Surya (FLAG license), Marker (FLAG license),

MinerU / PDF-Extract-Kit, RapidOCR.

- Table-structure recognition specifically (cell/grid reconstruction for borderless tables): TATR, PP-Structure, UniTable, etc.

Also assess: for SEMI-RULED / borderless bank tables, is a prompt-driven VLM (page image -> JSON transactions) more reliable than a classic OCR + table-structure pipeline? What are the accuracy/hallucination risks of VLMs on financial NUMBERS (transposed digits, dropped/duplicated rows), and how should output be VERIFIED? (muneem already machine-checks parses by balance reconciliation ? opening + sum(deposits) - sum(withdrawals) == closing, plus per-row running-balance walks and cross-checks against the bank's printed summary ? so any model's output can be automatically validated, which should inform the recommendation.)

Finally, briefly cover the best PAID/CLOUD fallbacks (clearly flagged approval-gated for sensitive data): AWS Textract (AnalyzeExpense / Tables), Google Document AI, Azure AI Document Intelligence, Reducto, LlamaParse, Anthropic/OpenAI vision ? note any self-hosting option and data-handling terms.

Deliver a RANKED shortlist of the most promising LOCAL options for muneem (NVIDIA GPU, Python, commercial-license-safe), each with license (code + weights), VRAM tier, maturity, strengths/weaknesses for dense bank-statement tables, and a concrete recommendation: which local model(s) to prototype for muneem's statement parsing and how they'd combine with the existing deterministic parsers + balance-reconciliation verification (e.g. VLM as a fallback/cross-check when deterministic parsing fails to reconcile). Cite sources and call out any license uncertainty explicitly.

## Claim under review

"The Surya CODE and the Surya WEIGHTS are licensed separately: the repo's code LICENSE file is Apache-2.0 (permissive), but the MODEL\_LICENSE explicitly excludes source code/scripts ('Complementary Material') from the OpenRAIL-M terms. Verifying the repo license alone would wrongly conclude Surya is fully permissive ? the weights are the binding constraint, and the current weights license is OpenRAIL-M, NOT the GPL/commercial dual-license that older descriptions cite."

**\*\*Source:\*\*** [https://github.com/datalab-to/surya/blob/master/MODEL\\_LICENSE](https://github.com/datalab-to/surya/blob/master/MODEL_LICENSE) (primary)

**\*\*Supporting quote:\*\*** "'Complementary Material' means the applicable source code and scripts used to define, run, load, benchmark or evaluate the Model, and used to prepare data for training or evaluation, if any."

## Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

**\*\*refuted=true\*\*** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

**\*\*refuted=false\*\*** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

## 2. assistant (2026-05-31T04:24:39.893Z)

I'll verify this claim about Surya's licensing by examining the primary source and checking the supporting quote.

## 3. assistant (2026-05-31T04:24:43.118Z)

I'll fetch the primary source (the MODEL\_LICENSE file) directly to verify the claim.

4. user (2026-05-31T04:24:49.612Z)

## License Information

**\*\*License Name:\*\*** "AI PUBS OPEN RAIL-M LICENSE (MODIFIED)"

**\*\*Does it define "Complementary Material"?\*\*** Yes. Verbatim definition:

"Complementary Material" means the applicable source code and scripts used to define, run, load, benchmark or evaluate the Model, and used to prepare data for training or evaluation, if any."

The license explicitly states: "Complementary Material is not licensed under this License."

**\*\*Is this OpenRAIL-M?\*** Yes, this is a modified version of OpenRAIL-M, as indicated in the title and preamble which states it is "generally applicable to any machine-learning Model."

5. user (2026-05-31T04:24:49.687Z)

## License Analysis

**\*\*License Name:\*\*** AI PUBS OPEN RAIL-M LICENSE (MODIFIED), Version 0.1, March 2, 2023

**\*\*Is this OpenRAIL-M?\*** Yes, the license explicitly states it is "generally applicable to any machine-learning Model" and uses the OpenRAIL-M framework with modifications.

**\*\*Definition of Complementary Material:\*\***

Yes, it is defined. Here's the verbatim definition:

"Complementary Material means the applicable source code and scripts used to define, run, load, benchmark or evaluate the Model, and used to prepare data for training or evaluation, if any."

**\*\*Exclusion from License Terms:\*\***

The definition explicitly excludes this material: "Complementary Material is not licensed under this License."

**\*\*What the License Covers vs. Excludes:\*\***

The license covers the machine-learning model itself and its derivatives. Notably, it excludes: complementary material (source code/scripts), accompanying data used in training, and documentation/tutorials. Section 1(d) also states: "The Data is not licensed under this License."

6. assistant (2026-05-31T04:24:52.575Z)

The primary source confirms the core claims. Let me verify the other parts: (1) the code LICENSE file is Apache-2.0, and (2) the older GPL/commercial dual-license characterization, to assess whether the claim is current and accurate.

7. user (2026-05-31T04:24:55.528Z)

## License Identification

This is the **\*\*Apache License 2.0\*\***.

The license header states verbatim:

"Apache License  
Version 2.0, January 2004  
<http://www.apache.org/licenses/>"

This is neither GPL nor another license?it is specifically Apache-2.0, a permissive open source license maintained by the Apache Software Foundation.

**8. user (2026-05-31T04:24:58.175Z)**

## Licensing Information from Surya README

**\*\*Code License:\*\*** Apache-2.0

The README states: "The Surya code is licensed under Apache 2.0."

**\*\*Model Weights License:\*\*** Modified AI Pubs Open Rail-M

The README specifies: "The model weights use a modified AI Pubs Open Rail-M license (free for research, personal use, and startups under \$5M funding/revenue)."

**\*\*Commercial Licensing:\*\***

For broader commercial use of the model weights, the README directs users to their pricing page, indicating commercial licensing is available through their managed platform at datalab.to.

The licensing is neither GPL nor standard OpenRAIL-M, but rather a modified version with specific restrictions on commercial applicability.

**9. assistant (2026-05-31T04:25:01.012Z)**

All key facts are now confirmed from primary sources. Let me do one cross-check on the "older GPL/commercial dual-license" historical characterization to confirm that part of the claim is accurate (that older descriptions cited GPL).

**10. user (2026-05-31T04:25:08.773Z)**

Web search results for query: "Surya OCR license GPL commercial dual license history weights"

Links: [{"title":"GitHub - CibiQuadance/surya-OCR: OCR and line detection in 90+ languages · GitHub","url":"https://github.com/CibiQuadance/surya-OCR"}, {"title":"GitHub - datalab-to/surya: OCR, layout analysis, reading order, table recognition in 90+ languages · GitHub","url":"https://github.com/datalab-to/surya"}, {"title":"pitapo/surya · Hugging Face","url":"https://huggingface.co/pitapo/surya"}, {"title":"GitHub - cognitiveailab/surya-sci: OCR, layout analysis, reading order, line detection in 90+ languages · GitHub","url":"https://github.com/cognitiveailab/surya-sci"}, {"title":"surya-ocr · PyPI","url":"https://pypi.org/project/surya-ocr/"}, {"title":"View as Markdown","url":"https://replicate.com/cudanexus/ocr-surya/llms.txt"}, {"title":"surya-ocr-vlite · PyPI","url":"https://pypi.org/project/surya-ocr-vlite/"}, {"title":"Surya: The Revolutionary OCR Toolkit for 90+ Languages","url":"https://converter.brightcoding.dev/blog/surya-the-revolutionary-ocr-toolkit-for-90-languages"}, {"title":"cudanexus/ocr-surya | Run with an API on Replicate","url":"https://replicate.com/cudanexus/ocr-surya"}, {"title":"surya/LICENSE at master · datalab-to/surya","url":"https://github.com/VikParuchuri/surya/blob/master/LICENSE"}]

Based on the search results, here's what I found regarding Surya OCR's licensing structure:

### Code License

The Surya code is licensed under Apache 2.0.

### Model Weights License

The licensing for model weights has evolved:

- **\*\*Earlier versions\*\***: The weights are licensed cc-by-nc-sa-4.0, but the license holder will waive that for any organization under \$5M USD in gross revenue in the most recent 12-month period.

- **\*\*Current version (Surya 2)\*\***: The model weights use a modified AI Pubs Open Rail-M license (free for research, personal use, and startups under \$5M funding/revenue).

## Dual Licensing for Commercial Use

If you want to remove the GPL license requirements (dual-license) and/or use the weights commercially over the revenue limit, commercial licensing options are available. For dual licensing options, you can contact the developers for removal of GPL license requirements for inference or commercial use over the revenue limit.

## Summary

Surya uses a dual licensing model where the code is open source (Apache 2.0), while the model weights allow free use for research, personal projects, and startups under a certain revenue threshold, with commercial licensing available for organizations exceeding those thresholds.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

### 11. assistant (2026-05-31T04:25:23.473Z)

I now have enough evidence to evaluate the claim thoroughly. Let me work through the verification.

**\*\*Verification against the checklist:\*\***

**\*\*1. Is the claim supported by the quote?\*\*** Yes, and more strongly than the quote alone shows. The primary source (MODEL\_LICENSE on master) is the "AI PUBS OPEN RAIL-M LICENSE (MODIFIED)". The supporting quote about "Complementary Material" is verbatim accurate. Critically, the license text explicitly states "Complementary Material is not licensed under this License" ? directly confirming the claim that source code/scripts are excluded from the OpenRAIL-M terms.

**\*\*2. Contradicting evidence?\*\*** None found. Independent confirmation across multiple sources:

- The repo's `LICENSE` file is verbatim "Apache License Version 2.0" (confirms code-license half).
- The README explicitly says: "The Surya code is licensed under Apache 2.0" and "The model weights use a modified AI Pubs Open Rail-M license."
- So the code/weights split is real and confirmed from two independent files plus README.

**\*\*3. Source quality?\*\*** Primary source (the actual license files on the canonical `datalab-to/surya` repo on master branch). Highest possible quality for a license claim.

**\*\*4. Outdated?\*\*** This is the strongest part of the claim and it checks out. The claim asserts the *current* weights license is OpenRAIL-M, NOT the older GPL/CC-BY-NC dual-license. My search confirms the license **\*\*evolved\*\***: earlier Surya versions used CC-BY-NC-SA-4.0 / GPL-dual-license framing; the current version (Surya 2) uses the modified AI Pubs OpenRAIL-M. The claim correctly identifies the current state and correctly notes that older descriptions cite the GPL/commercial dual-license. This is exactly right.

**\*\*5. Marketing/cherry-picked?\*\*** No ? these are legal license texts, not benchmarks or marketing.

**\*\*One minor nuance:\*\*** The claim says verifying the repo license alone "would wrongly conclude Surya is fully permissive." This is the load-bearing analytical point, and it is correct ? the Apache-2.0 LICENSE file governs only the code; the weights are bound by OpenRAIL-M (free only for research/personal/startups under \$5M, commercial license required otherwise). For muneem (a personal-use tool), the weights are actually usable under the personal-use carve-out, but the claim's *licensing characterization* (weights ? permissive; repo-license-only check is misleading) is accurate regardless. The claim is about license classification, not about muneem's specific eligibility, so this doesn't undercut it.

Every component of the claim is supported by primary sources and is current.

### 12. user (2026-05-31T04:25:34.142Z)

Structured output provided successfully

### 13. assistant (2026-05-31T04:25:43.008Z)

Claim verified as accurate. I attempted to refute it but could not ? every component is confirmed by primary sources (the actual LICENSE and MODEL\_LICENSE files on the canonical `datalab-to/surya` repo, master branch), with independent corroboration from the README and web search.

**\*\*Verdict: refuted=false (claim stands), high confidence.\*\***

#### Key findings:

- **\*\*Code LICENSE\*\*** = verbatim "Apache License Version 2.0" (confirms permissive code half).
- **\*\*MODEL\_LICENSE\*\*** = "AI PUBS OPEN RAIL-M LICENSE (MODIFIED), Version 0.1" ? confirms the weights are OpenRAIL-M, separately licensed.
- **\*\*Supporting quote is verbatim accurate\*\***, and the license goes further: "Complementary Material is not licensed under this License" ? directly confirming source code/scripts are carved out of the OpenRAIL-M terms.
- **\*\*Currency check (the claim's strongest assertion) holds\*\***: the license evolved. Older Surya used CC-BY-NC-SA-4.0 / GPL-dual-license framing; current Surya (2) uses the modified AI Pubs OpenRAIL-M. The claim correctly identifies the current binding constraint and correctly notes older descriptions cite the GPL/commercial dual-license.

The only nuance worth flagging for the parent report: the OpenRAIL-M weights are free for "research, personal use, and startups under \$5M," so for muneem (a personal-use tool) the weights would technically be *usable* under the personal-use carve-out ? but the claim is about license *classification* (weights are restrictive, not MIT/BSD/Apache, and a repo-license-only check is misleading), which is fully accurate.

#### Sources:

- [Surya MODEL\_LICENSE (master)]([https://github.com/datalab-to/surya/blob/master/MODEL\\_LICENSE](https://github.com/datalab-to/surya/blob/master/MODEL_LICENSE))
- [Surya LICENSE (master)](<https://raw.githubusercontent.com/datalab-to/surya/master/LICENSE>)
- [datalab-to/surya README](<https://github.com/datalab-to/surya>)